Senior Researcher
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Daejeon , KR

Wookyoung Kim

19 papers SCI + KCI peer-reviewed	18 patents domestic + 1 US	8 programs registered software	5 transfers technology transfers

Summary

Thermal engineer specializing in data center cooling, heat exchangers, heat pump systems, and two-phase heat transfer. Experienced in both experimental research and engineering software development.

Education

B.S. · KAIST · Mechanical Engineering — 2011 – 2015

M.S. · KAIST · Mechanical Engineering — 2015 – 2017

· Thesis: Effect of artificial cavities on the thermal performance of pulsating heat pipes
· Advisor: Prof. Sung Jin Kim

Ph.D. · KAIST (Korea Advanced Institute of Science and Technology) · Mechanical Engineering — 2017 – 2021

· Dissertation: Study on a Thermal Network-based Model for Predicting the Thermal Resistance of Pulsating Heat Pipes
· Advisor: Prof. Sung Jin Kim

Experience

2021 – Present

Senior Researcher · Korea Institute of Machinery and Materials (KIMM) · Daejeon, Republic of Korea — 2021 – Present

Heat Pump Research Center, Department of Thermal Energy Solutions, Innovative Energy Machinery Research Division

- PI: Jet-enhanced immersion cooling for next-gen data center cooling
- PI: KIMMPROP - Thermophysical property iOS/Android app development
- Lead: PCHE design & testing for liquid hydrogen vaporizers (down to -220°C, 100 MPa)
- Lead: Immersion cooling waste heat utilization and thermal management
- 19 journal papers (11 SCI + 8 KCI), 18 domestic patents, 1 U.S. patent, 8 registered software programs, 5 technology transfers

Publications

11 SCI · 8 KCI · 11 Conference

SCI / International Journal Articles 11 papers

2026 — Falling film evaporation of ammonia on smooth and Low-Fin tube arrays
Thermal Science and Engineering Progress
Kim, Jin Sub and Kim, Wookyoung and Kim, Hak Soo and Kim, Young

2025 — Experimental and theoretical investigation of freezing phenomenon in Printed Circuit Heat Exchanger
Applied Thermal Engineering
Kim, Bogyeom and Lee, Kong Hoon and Kim, Minchang and Moon, Sunyoung and Yoo, Jin Woo and Kim, Wookyoung

2024 — Experimental studies on the VLE of R-32/R-125 and verification of experimental setup
Journal of Mechanical Science and Technology
Kim, Dong Ho and Kim, Wookyoung and Kim, Jeong Hyun and Kim, Hak Soo and Lee, Kong Hoon

2024 A numerical study on the performance of chemisorption heat pump according to various design conditions

Applied Thermal Engineering

Kim, Hak Soo and Kim, Dong Ho and Kim, Jin Sub and Kim, Wookyoung and Kim, Young

2024 Pool boiling heat transfer of ammonia outside enhanced tubes with various fin structures

Applied Thermal Engineering

Kim, Jin Sub and Kim, Wookyoung and Kim, Hak Soo and Kim, Young

2024 Experimental study of a chemisorption heat pump under different operation conditions

Applied Thermal Engineering

Kim, Hak Soo and Kim, Jeong Hyun and Kim, Jin Sub and Kim, Wookyoung and Kim, Young

2023 Characterization of liquid behavior in distributor of falling-film evaporator

Physics of Fluids

Kim, Jungchul and Kim, Chan Geun and Kim, Hak Soo and Kim, Wookyoung and Kim, Dong Ho

2023 Pool boiling heat transfer of ammonia outside a single tube with fin structures: Hysteresis phenomena and boiling enhancement

International Communications in Heat and Mass Transfer

Kim, Jin Sub and Kim, Wookyoung and Kim, Hak Soo and Kim, Young and Yoon, Seok Ho

2021 Fundamental issues and technical problems about pulsating heat pipes

Journal of Heat Transfer - Transactions of the ASME

Kim, Wookyoung and Kim, Sung Jin

2020 Effect of a flow behavior on the thermal performance of closed-loop and closed-end pulsating heat pipes

International Journal of Heat and Mass Transfer

Kim, Wookyoung and Kim, Sung Jin

2018 Effect of reentrant cavities on the thermal performance of a pulsating heat pipe

Applied Thermal Engineering

Kim, Wookyoung and Kim, Sung Jin

KCI / Korean Journal Articles 8 papers

2026 A Study on the Design of an Air-to-Air Plate Heat Exchanger for 300°C High Temperature Heat Pumps

Korean Journal of Air-Conditioning and Refrigeration Engineering

Kim, Jin Sub and Shin, Dong Hwan and Shim, Jaechwan and Lee, Kong Hoon and Sohn, Sangho and Kim, Wookyoung and Song, Chanho

2024 Experimental Investigation on the Freezing Condition of Printed Circuit Heat Exchanger for Cryogenic Liquid Hydrogen Vaporizer

Journal of Hydrogen and New Energy

Kim, Wookyoung and Kim, Bokyem and Sohn, Sangho and Lee, Kong Hoon and Kim, Jungchul

2024 An Experimental Study on The Thermal Performance Measurement and Flow Visualization of Falling Film Evaporator Using R-1233ZD(E) Refrigerant

Korean Journal of Air-Conditioning and Refrigeration Engineering

Kim, Wookyoung and Kim, Hak Soo and Kim, Jungchul and Kim, Dong Ho

2023 An Experimental Study on the Performance Characteristics of a Falling Film Evaporator for R-1234ze(E) Refrigerant

Korean Journal of Air-Conditioning and Refrigeration Engineering

Kim, Dong Ho and Kim, Hak Soo and Kim, Jungchul and Kim, Wookyoung

2023 Comparative Study of Air Cooling and Immersion Cooling for the Thermal Management of a Cylindrical Battery Pack

Transactions of the Korean Society of Mechanical Engineers B

Kim, Jin Sub and Kim, Seul Ah and Shin, Dong Hwan and Kim, Wookyoung and Moon, Sunyoung and Chung, Yoong and Sohn, Sangho

2022 A Study on Anti-Icing Design by Conjugate Heat Transfer Analysis in a Lab-Scale Printed Circuit Heat Exchanger for Supply of Cryogenic High Pressure Liquid Hydrogen
Transactions of the Korean Hydrogen and New Energy Society
Sohn, Sangho and Kim, Wookyoung

2022 Numerical Study on the Performance of Hybrid Falling Film Evaporator
Korean Journal of Air-Conditioning and Refrigeration Engineering
Kim, Hak Soo and Kim, Wookyoung and Lee, Kong Hoon and Kim, Dong Ho

2022 Selection of the Equation of State for Thermodynamic Properties in the Low GWP Mixture Refrigerants
Transactions of the Korean Society of Mechanical Engineers B
Lee, Kong Hoon and Kim, Dong Ho and Kim, Wookyoung and Kim, Jin Sub and Kim, Young and Yoo, Jin Woo

Conference Proceedings `11 papers`

2024 Experimental and theoretical investigation on avoiding freezing phenomena of Printed Circuit Heat Exchanger for cryogenic liquid hydrogen vaporizer
Proceedings of the 29th International Cryogenic Engineering Conference (ICEC29)
Kim, Wookyoung and others

2024 Understanding the thermal characteristics of falling film evaporation of R-1233ZD(E) refrigerant using flow visualization and heat transfer measurement
Proceedings of the 11th Asian Conference on Refrigeration and Air-conditioning (ACRA)
Kim, Wookyoung and others

2023 Experimental investigation on the flow and thermal characteristics of falling film evaporator using R-1233ZD(E) refrigerant
Proceedings of the 11th International Conference on Boiling and Condensation Heat Transfer (ICBCHT)
Kim, Wookyoung and others

2019 Experimental Investigation on the Thermal Performance of PHP Operating in a Circulation/oscillation Mode
Proceedings of the Korea-China-Japan Joint Symposium
Kim, Wookyoung and Kim, Sung Jin

2019 Comparison of thermal performance between CEPHP and CLPHP
Proceedings of the 4th Thermal and Fluids Engineering Conference (TFEC)
Kim, Wookyoung and Kim, Sung Jin

2018 Experimental investigation on the thermal performance of double-condenser pulsating heat pipes
Proceedings of the 10th International Conference on Boiling and Condensation Heat Transfer (ICBCHT)
Kim, Wookyoung and Kim, Sung Jin

2018 Development of deployable radiator for spacecraft using a pulsating heat pipe
Proceedings of the KSME 2018 Spring Annual Meeting (Thermal Engineering Division)
Kim, Wookyoung and Kim, Sung Jin

2018 Effects of circulating flow on the thermal performance of a pulsating heat pipe
Proceedings of the KSME 2018 Annual Meeting
Kim, Wookyoung and Kim, Sung Jin

2017 Experimental Investigation on the Thermal Performance of a Pulsating Heat Pipe with Artificial Cavities
Proceedings of the 1st Asian Conference on Thermal Sciences (ACTS 2017)
Kim, Wookyoung and Kim, Sung Jin

2017 Experimental investigation of artificial cavities on the thermal performance of a pulsating heat pipe
Proceedings of the ASME InterPACK 2017
Kim, Wookyoung and Kim, Sung Jin

2016 Effects of artificial cavities on the startup performance of a pulsating heat pipe
Proceedings of the KSME 2016 Annual Meeting
Kim, Wookyoung and Kim, Sung Jin

Research Projects

25 entries

Development of an AI-based Automated Design Tool for Data Center Direct Liquid Cooling (DLC) Systems

2026

AI-driven automated design tool for data center direct liquid cooling (DLC) systems, using surrogate modeling and optimization for cold-plate and cooling-system design (PI)

Development of a Direct Liquid Cooling (DLC) System for Reducing Power Consumption in Data Centers

2026 – 2029

National R&D project (KETEP, Ministry of Climate, Energy and Environment) to develop a direct liquid cooling (DLC) system that reduces data center power consumption (Lead)

Jet-Enhanced Immersion Cooling for Data Centers

2025

Development of high-efficiency jet-impingement immersion cooling technology for next-generation high-heat-density data center servers (PI)

KIMMPROP - Thermophysical Property iOS/Android App

2025

Cross-platform iOS/Android app for thermal-fluid property calculations using CoolProp/REFPROP via WASM (PI)

PCHE for Liquid Hydrogen Supply System

2021 – 2026

Core material development for liquid hydrogen supply system - PCHE performance testing and design (Lead)

Compact PCHE Design for Liquid Hydrogen

2022 – 2026

Compact heat exchanger (PCHE) design technology for liquid hydrogen below -200°C, 100 MPa class (Lead)

Immersion Cooling Waste Heat Utilization

2024 – 2028

Immersion cooling waste heat utilization and active thermal management technology (Lead)

300°C High-Temperature Heat Pump System

2023 – 2028

300°C-class high-temperature heat pump system for fossil fuel replacement (Participant)

Chemisorption Heat Pump with Electrochemical Compressor

2021 – 2025

Chemisorption heat pump system using electrochemical compressor (Participant)

Next-Gen Alternative Refrigerant & HVAC Technology

2021 – 2025

Next-generation alternative refrigerant and high-efficiency HVAC core technology development (Participant)

Low-Charge Industrial Heat Pump

2022 – 2023

Industrial heat pump with low refrigerant charge and simultaneous hot/cold water production (Participant)

Water-Source Heating/Cooling Hybrid System

2022 – 2023

Water-source heating/cooling and regenerative heat hybrid system (Participant)

Heat Pipe HX Performance Enhancement

2022

Heat pipe heat exchanger performance enhancement using boiling (Participant)

Post H-class Gas Turbine Technology

2022

Post H-class gas turbine technology development (Participant)

Compact Inverter Compressor Dress Care System

2022

Compact inverter compressor-based dress care system development (Participant)

Ultra-Thin Vapor Chamber Performance Enhancement
2022 – 2023

Performance enhancement technology for ultra-thin vapor chambers for semiconductor cooling (Participant)

PCHE Freezing Test for Vehicle Heat Exchangers
2025

PCHE freezing performance test for automotive heat exchanger applications (PI)

BTMS Integrated A/Con System Evaluation
2025

Battery thermal management system integrated A/Con system real-vehicle evaluation process development (Participant)

3D Multiporous Cooling System for Ultra-high Heat Flux Applications
2021

Development of a 3D multi-porous cooling device for ultra-high-heat-flux heat-generating elements (Graduate Researcher)

Smart protective suit with active cooling/ventilation and antivirus suit changing system
2020

Active cooling and ventilation system for a smart protective suit enabling on-body thermal management (Graduate Researcher)

Next-Generation CPI Reliability Structural & Thermal Design
2020

Structural and thermal design technology to secure the reliability of next-generation CPI; NRF original-technology development project (Graduate Researcher)

OHP High-Pressure Working-Fluid Charging Technology
2020

Technical service for high-pressure working-fluid charging of oscillating heat pipes (OHP); commissioned by Korea Institute of Energy Research (Graduate Researcher)

Thermal optimization of MMU rear housing for reducing the weight
2019 – 2020

Heat-dissipation performance optimization and weight reduction of an MMU rear housing; industry project funded by Samsung Electronics (Graduate Researcher)

Development of a Flexible Thin-Film Ultra-Thermal-Conductor
2015 – 2020

NRF basic research on flexible thin-film ultra-high-thermal-conductivity heat spreaders based on pulsating heat pipes (Graduate Researcher)

Study on a smart space thermal management system
2016 – 2018

Pulsating-heat-pipe-based deployable radiator and smart thermal control device for spacecraft; contract with Korea Aerospace University (Graduate Researcher)

Patents & Registered Software

Patents `19 registered`

2023	Heat exchanger with anti-freezing capability (1)
2023	Heat exchanger with anti-freezing capability (2)
2024	Modular fin-tube reactor
2024	Fin-tube type reactor
2022	Tray structure for heat exchanger
2022	Oil recovery system for refrigeration systems (1)
2022	Oil recovery system for refrigeration systems (2)
2023	Micro-channel reactor
2024	Immersion cooling device
2023	Geothermal heat supply device and heating system

2023	Heat exchanger thermal performance testing apparatus
2022	Ternary refrigerant composition and heat pump system
2023	Display device
2022	Duct assembly and combustor
2022	Immersion cooling HVAC system and method
2022	Heat pipe integrated reactor for adsorption heat pump
2023	Adsorption heat pump evaporator and system
2018	Plate pulsating heat spreader with artificial cavities
2019	Plate pulsating heat spreader with artificial cavities (US 10,436,520 B2)

Registered Software `8 programs`

2024	PCHE thickness design program (C-2024-052179)
2024	High-temperature heat pump cycle design program (C-2024-038691)
2023	Heat exchanger HTC uncertainty analysis program (C-2023-059071)
2023	Lab-scale heat exchanger analysis program (C-2023-056463)
2025	100 kg/hr hydrogen PCHE design program (C-2025-056891)
2025	Adsorption cooling simulation program (C-2025-034652)
2024	Vapor chamber performance analysis program (C-2024-041200)
2024	Fin-tube heat exchanger design program (C-2024-038692)

Core Skills

EXPERIMENTAL	Two-phase heat transfer testing · Cryogenic systems (down to -220°C) · High-pressure testing (up to 100 MPa) · PCHE diffusion bonding & fabrication · Immersion cooling (CPU/GPU servers) · Flow visualization (high-speed camera) · VLE measurement & gas chromatography
ANALYTICAL & COMPUTATIONAL	Thermal network modeling · Heat exchanger design (PCHE, S&T, PHE) · CFD simulation · Machine learning & surrogate modeling (scikit-learn, PyTorch, LightGBM) · CoolProp/REFPROP
SOFTWARE DEVELOPMENT	Python, JavaScript, TypeScript · FastAPI, React Native · Git/GitHub, Linux, CMake · Cross-platform app development (iOS/Android) · AI/LLM agent development

Research Interests

Data Center Cooling
Immersion cooling · Direct liquid cooling · Jet impingement · PUE optimization

High-Heat-Flux Electronics Cooling
CPU/GPU cold plate · Focused cooling · Next-gen chip thermal management

Heat Exchangers
PCHE · Shell & Tube · Plate HX · Cryogenic applications

Heat Pump Systems
Chemisorption · Adsorption · High-temperature (300°C)

Low GWP Refrigerant Systems
Eco-friendly refrigerants · VLE measurement · Equation of state development

Two-Phase Heat Transfer
Boiling · Evaporation · Pulsating heat pipes · Vapor chambers

Cryogenic Heat Transfer
Liquid hydrogen vaporizer · Supercritical fluid · Sub-200°C systems

Professional Affiliations

Council Member · Korean Society of Mechanical Engineers (KSME) 2026-04 – Present

Committee Member - Thermal Management / Cooling · SAREK Data Center Technology Committee 2026-03 – Present

Regular Member · Society of Air-Conditioning and Refrigerating Engineers of Korea (SAREK) 2022-04 – Present

Regular Member · Korean Society of Mechanical Engineers (KSME) 2021-05 – Present

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